

CURRICULUM VITAE

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Research Interests

Artificial Intelligence & Machine Learning, Natural Language Processing, Computer Vision, Blockchain Systems, Distributed Computing, Intelligent Scheduling, Optimization Algorithms, Accessibility Technology, Indian Knowledge Systems

My research sits at the intersection of **optimization algorithms** and **real-world systems**. I am particularly interested in applying multi-objective optimization techniques — such as genetic algorithms and evolutionary computation — to complex urban-scale problems where competing objectives must be balanced. My current work explores how probabilistic pre-analysis and machine learning can enhance operating system scheduling, and how unified multi-objective formulations can improve urban mobility systems. I am also passionate about building AI-powered accessibility tools that serve underserved communities, combining natural language processing, computer vision, and human-computer interaction to create inclusive technology.

Academic Qualifications

No.	Degree / Examination	Institution	Duration	Result
1	B.Tech in Computer Science & Engineering	Amrita Vishwa Vidyapeetham, Bengaluru	2024 – 2028	CGPA: 7.86

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Artificial Intelligence, Machine Learning, Software Engineering, Cloud Computing, Distributed Systems, Cryptography & Network Security.

Academic Activities: Active member of Google Developer Student Club (GDSC); participant in multiple hackathons and technical fests. Completed research projects in multi-objective optimization and intelligent scheduling under the guidance of faculty at Amrita Vishwa Vidyapeetham.

Publications & Research

1. ORCA — Optimized Routing & Car-park Assignment via Multi-Objective Genetic Algorithms

Lohitaksha Patary, Aditi Singhal, Nithilan Rameshkumar, Murali K.

DZone, 2025

Abstract: This paper introduces ORCA (Optimized Routing Car-park Assignment), a unified multi-objective optimization framework that jointly solves vehicle routing and parking allocation as a single connected problem — a gap not addressed by prior work. Existing Parking Guidance Systems (PGS) treat routing and allocation sequentially, failing to capture the feedback loop where parking choices affect network-wide congestion. ORCA models the city as a dynamic weighted graph and employs a Multi-Objective Genetic Algorithm (MOGA) to discover Pareto-optimal solutions that simultaneously minimize four competing objectives: driving time (f_{drive}), walking distance (f_{walk}), parking cost (f_{cost}), and system-wide congestion (f_{cong}).

Key Contributions:

- A unified multi-objective formulation capturing all four trade-offs in a single fitness function with binary decision variables x_{ij} and capacity constraints.

- ▶ A custom MOGA with smart population seeding (20% FCFS, 20% spread, 60% random), capacity-aware mutation, single-point crossover, and Pareto-based tournament selection with weighted tie-breaking.
- ▶ Traffic modeling via the Bureau of Public Roads (BPR) function: $T_e = T_e^{\text{free}} \cdot \left[1 + 0.15 \left(\frac{V_e}{C_e} \right)^4 \right]$, with congestion measured as normalized mean squared volume-to-capacity ratios plus bottleneck penalties.
- ▶ Three-tier deployment architecture (IoT sensing → Edge aggregation → Cloud optimization) for real-world smart city integration.

Evaluation: Validated on real road network data from Koramangala, Bengaluru (3,865 nodes, 5,173 edges, 80 parking spots, 195 vehicles). ORCA achieves 14.8% lower walking distance than FCFS while maintaining competitive driving times. Monte Carlo simulations with Kruskal-Wallis and Mann-Whitney U-tests confirm statistical significance ($p < 0.001$). The framework generates a diverse Pareto front enabling dynamic policy selection based on real-time conditions.

Technologies: Python, NetworkX, OSMnx, Matplotlib, Folium, Streamlit. Paper written in Typst (IEEE format). Source code and paper hosted on GitHub.

Impact: Addresses UN Sustainable Development Goals 11 (Sustainable Cities & Communities) and 9 (Industry, Innovation & Infrastructure). Provides a deployable blueprint for next-generation smart parking systems that balance individual driver convenience with global network health.

2. PRISM — Probabilistic Real-time Intelligent Scheduling Mechanism

Lohitaksha Patary

Target venue: IEEE ICCA / ICCCS

Abstract: PRISM presents a three-layer hybrid intelligent scheduling framework that addresses the fundamental limitation of traditional OS schedulers — their reactive nature. Conventional algorithms like Round Robin, FCFS, and SJF make no predictions about process behavior before execution, resulting in unnecessary context switches, cache misses, and higher turnaround times. PRISM introduces probabilistic pre-analysis and machine learning to predict process characteristics before scheduling decisions are made, minimizing context switching overhead and improving system throughput.

Architecture — Three-Layer Framework:

- ▶ **Layer 1 — Probabilistic Pre-Analysis:** Uses Markov Chains to model state transition probabilities between CPU and I/O bursts. Poisson arrival rate modeling captures the statistical distribution of process arrivals. This layer predicts the expected burst pattern for each process before it enters the ready queue, enabling proactive scheduling decisions rather than reactive ones.
- ▶ **Layer 2 — ML Classification:** A Random Forest classifier trained on process behavior features (historical burst patterns, arrival statistics, resource usage) categorizes incoming processes into CPU-bound, I/O-bound, or Mixed types. This classification enables the scheduler to apply type-specific scheduling policies — for example, prioritizing I/O-bound processes to keep devices busy while CPU-bound processes wait.
- ▶ **Layer 3 — Sequence Prediction:** A Weighted Moving Average (WMA) model forecasts future burst sequences based on historical patterns. This enables the scheduler to predict when a process will likely block for I/O, allowing preemptive decisions that reduce idle CPU time and context switch overhead.

Online Learning: An adaptive feedback loop continuously updates the Markov transition matrix and Random Forest model parameters based on observed process behavior. This ensures the scheduler improves its predictions over time, adapting to changing workload characteristics without manual retuning.

Evaluation: Benchmarked against FCFS, SJF, Round Robin (quantum = 4ms), and Priority Scheduling using five metrics: Average Turnaround Time, Average Waiting Time, Context Switch Count, CPU Utilization (%), and Jain's Fairness Index. Results demonstrate that PRISM reduces context switches by leveraging predictive pre-analysis, achieving superior turnaround and waiting times while maintaining fairness across all process types.

Technologies: Python 3.x, NumPy, Scikit-learn (Random Forest), Matplotlib. Custom discrete-event simulation loop with pure queue control representation (no external simulation frameworks). Source code available on GitHub.

3. How MOGA Solves the Urban Parking Assignment Problem: Inside the ORCA System

Lohitaksha Patary

DZone, 2025

A technical article published on DZone explaining the ORCA framework to a broader engineering audience. The article details how Multi-Objective Genetic Algorithms can be applied to urban parking optimization, covering the graph modeling approach, objective function design, GA parameter tuning, and practical deployment considerations. It bridges the gap between the academic paper and industry practitioners, making the research accessible to software engineers and smart city developers.

Work Experience

A. Research & Development Experience

1. **Computer Vision Intern — Cellium Networks** *Oct 2025 – Present Remote (US-based company)*
 - Stitch multi-camera vehicle feeds into synchronized panoramic BEV (Bird's Eye View) output using OpenCV, homography transformations, and feature matching algorithms.
 - Applied YOLO-based object detection on stitched BEV for real-time obstacle identification in surround-view perception systems, enabling autonomous driving applications.
 - Implemented Kalman filter-based multi-object tracking across camera frames to maintain consistent object identities during cross-camera transitions.
 - Developed calibration pipelines for extrinsic camera parameter estimation, reducing stitching artifacts by 35% compared to naive concatenation approaches.
2. **AI/Accessibility Intern — IBM SkillsBuild AI Internship** *Nov 2025 – Jan 2026 Remote*
 - Built **Granite Accessible Assistant** — a full-stack RAG-powered accessibility kiosk AI using IBM Granite 4.0 Micro as the LLM backbone, FAISS vector search for semantic retrieval, and SentenceTransformers for embedding generation. Designed to serve users with visual, auditory, and cognitive disabilities.
 - Designed a profile-driven prompt engineering layer that dynamically adapts responses based on disability type (visual, auditory, cognitive) and user language preferences, incorporating few-shot examples and system personas for each accessibility profile.
 - Integrated BLIP image captioning for scene understanding and MediaPipe hand gesture recognition for touchless interaction, enabling users with motor disabilities to interact without physical contact.
 - Engineered the complete RAG pipeline covering document chunking, embedding generation, vector storage, similarity search, and context-aware response generation with source attribution.
 - Conducted accessibility testing with screen readers and keyboard-only navigation to ensure WCAG 2.1 AA compliance across the entire user interface.
3. **Cloud Engineer — GDG Amrita Vishwa Vidyapeetham** *Sept 2024 – Present Bengaluru, India*
 - Lead study jams and hands-on cloud workshops for peers, preparing cloud-native sample projects and deployment guides on Google Cloud Platform.
 - Mentor fellow students on cloud architecture, API design, and modern web deployment through practical demos and code reviews.
 - Organize weekly technical sessions covering containerization, serverless computing, and CI/CD pipelines with hands-on lab components.
4. **Event Organizer — Dastaan Tech Fest 2024** *Oct 2024 Bengaluru, India*
 - Served as one of the primary hosts for IBM's cloud-native language workshop at Dastaan Tech Fest, Amrita's flagship technical festival.
 - Coordinated end-to-end logistics including speaker liaison, venue setup, participant registration, and live session moderation for 80+ attendees.
 - Facilitated hands-on coding segments during the workshop, assisting participants with cloud deployment exercises and troubleshooting in real time.

B. Research Projects

1. **ORCA — Optimized Routing & Car-park Assignment** *Jan 2025 – Present*
 - Research project applying Multi-Objective Genetic Algorithms on real OSM street networks to assign parking while balancing travel distance, congestion, and sensor cost.
 - Modeled city as dynamic weighted graph with BPR traffic function for congestion-aware routing; implemented capacity-aware mutation and smart population seeding in the GA.
 - Benchmarked on Koramangala, Bengaluru (3,865 nodes, 80 parking spots) against FCFS and Weighted Greedy baselines with Monte Carlo simulation and statistical significance testing.
 - Written in Typst (IEEE format); paper and source code openly hosted on GitHub. Targets UN SDGs 11 and 9.
2. **PRISM — Probabilistic Real-time Intelligent Scheduling Mechanism** *2025 – 2026*
 - Three-layer hybrid framework: Layer 1 — probabilistic pre-analysis using Markov Chains and Poisson arrival rates; Layer 2 — ML classification using Random Forest (CPU-bound vs I/O-bound vs Mixed); Layer 3 — sequence prediction using Weighted Moving Average for burst forecasting.
 - Online learning feedback loop that continuously improves predictions based on observed process behavior.
 - Benchmarked against FCFS, SJF, Round Robin (quantum = 4ms), and Priority Scheduling on five metrics: Average Turnaround Time, Average Waiting Time, Context Switch Count, CPU Utilization, Jain's Fairness Index.
 - Target venue: IEEE ICCA / ICCCS. Source code available on GitHub.
3. **IntelliQuery — Natural Language SQL Engine** *Sept 2024 – Dec 2024*
 - Java service that translates natural-language questions into optimized SQL and explains results for non-technical teams.
 - Integrated LLM pipeline translating questions into SQL executed on MariaDB via JDBC.
 - Designed modular architecture enabling conversational access to enterprise datasets.

Software Projects

1. **PharmaChain — Blockchain Medicine Tracker** *Apr 2024 – Present* Blockchain-powered supply chain visibility platform for pharmaceutical batches.
 - Built Rust backend services exposing REST APIs to a React + TypeScript dashboard.
 - Added role-aware batch provenance and medicine history tracking for hospitals.
 - Prototyping Merkle tree verification to enable tamper-evident auditing.
 - Designed role-based access control for hospital administrators, pharmacists, and regulatory auditors.
2. **Granite Accessible Assistant — IBM** *Nov 2025 – Jan 2026*
 - A full-stack RAG-powered accessibility kiosk AI using IBM Granite 4.0 Micro, FAISS vector search, and SentenceTransformers.
 - Serves users with visual, auditory, and cognitive disabilities through multi-modal interaction.
 - Integrates BLIP vision captioning and MediaPipe gesture recognition for touchless operation.
 - Deployed as a Streamlit-based web application with responsive design for various screen sizes and input modalities.
3. **Campus Carpool Platform** *July 2024 – Present* Ride-sharing web app for students and faculty built with Next.js and Supabase.
 - Designed Supabase data model with row-level security for rider and driver info.
 - Added ride pinging, time-based matching, and notifications via Edge Functions and realtime.
 - Implemented geolocation-based matching algorithm considering route overlap and schedule compatibility.
4. **College Canteen Digitization System** *Feb 2024 – June 2024* Team-led effort to digitize campus canteen ordering with end-to-end web access.
 - Co-developed ordering, menu, and staff dashboards using Node.js, Express, MongoDB.
 - Automated queue updates and payment confirmations to reduce wait times.
 - Built real-time order tracking system with push notifications for order status changes.
5. **HandSlash — Touchless Hand Gesture Recognition System** *2025* AI-powered touchless interaction system using real-time hand landmark detection for camera-based gesture control.
 - Built real-time hand tracking pipeline using MediaPipe Tasks HandLandmarker with TFLite float16 model bundle, achieving sub-30ms inference on consumer hardware.
 - Implemented gesture classification for grab, click, and type actions by analyzing 21 hand landmark coordinates per frame with custom heuristics for finger extension and curl detection.
 - Designed a touchless UI layer enabling users to interact with digital interfaces without physical contact — applicable to accessibility kiosks, sterile environments, and public displays.
 - Won **First Place** at Edge-Core Hackathon among 25 competing teams for technical innovation and clean implementation.

Certifications & Professional Development

- Feb 2026** Blockchain Opportunity Analysis — INSEAD (Coursera)
8-course professional certification on blockchain strategy and opportunity evaluation, completed under INSEAD faculty and the Blockchain Research Institute. Capstone: PharmaChain — a blockchain-powered medicine supply chain tracker. Topics covered: blockchain fundamentals, tokenization, smart contracts, DeFi, DAOs, and enterprise blockchain strategy.
- 2025** IBM SkillsBuild AI Internship Certificate
Completed AI internship focused on RAG systems, accessibility technologies, and LLM deployment. Built Granite Accessible Assistant using IBM Granite 4.0 Micro. Gained hands-on experience with FAISS, SentenceTransformers, BLIP, and MediaPipe.
- 2025** Google AI/ML Certification (Coursera)
Foundations of machine learning, deep learning, and natural language processing. Covered supervised/unsupervised learning, neural networks, CNNs, RNNs, and transformer architectures.

Technical Skills

Programming Languages	Python, Java, Rust, TypeScript/JavaScript, Solidity, Solana (Rust)
AI / Machine Learning	Large Language Models, RAG, Computer Vision, Deep Learning, YOLO, OpenCV, FAISS, SentenceTransformers, Scikit-learn, Random Forests, Markov Chains
Web & Backend	React, Next.js, Node.js, Express, Spring Boot, REST APIs, JDBC, Supabase
Databases & Cloud	MongoDB, PostgreSQL, MariaDB, Google Cloud Platform, Docker, Edge Functions
Blockchain	Ethereum, Solana (Rust), Merkle Trees, Smart Contracts, Supply Chain Applications
DevOps & Tools	Docker, Git, Typst, Figma, Matplotlib, Folium, Streamlit, LaTeX
Research Methods	Multi-Objective Optimization, Genetic Algorithms (NSGA-II), Discrete-Event Simulation, Markov Chains, Statistical Significance Testing (Kruskal-Wallis, Mann-Whitney U), IEEE/ACM Formatting

Languages Known

English Fluent (Read, Write, Speak)
French Intermediate (Read, Write, Speak)
Bengali Native
Hindi Fluent
German Beginner

Extracurricular & Achievements

- **Piano Grade 5** — Trinity College London Certification; active composer and performer
- **Regional Art Champion** — Global Art Competition (Third Runner-up at national level)
- **High School Piano Competition** — Second Place in inter-school piano competition
- **Edge-Core Hackathon** — First Place; built HandSlash touchless hand gesture recognition system using MediaPipe HandLandmarker
- **GDG Active Member** — Lead study jams and cloud workshops at Amrita
- **Dastaan Tech Fest Host** — Coordinated IBM's cloud-native workshop for 80+ attendees

Teaching & Community Involvement

Technical Mentoring

- ▶ **GDG Cloud Workshops (Sept 2024 – Present):** Designed and delivered hands-on workshops on Google Cloud Platform for computer science students. Topics covered include containerization with Docker, serverless computing with Cloud Functions, and CI/CD pipelines with Cloud Build. Prepared cloud-native sample projects and deployment guides used as reference material by workshop participants.
- ▶ **Code Review & Open Source:** Actively review pull requests on collaborative GitHub projects. Contributed bug fixes and feature implementations to open-source repositories in the fields of web development and machine learning.

Hackathon Leadership

- ▶ **Dastaan Tech Fest 2024 (Oct 2024):** Served as one of the primary hosts for IBM's cloud-native language workshop at Amrita's flagship technical festival. Managed end-to-end logistics including speaker liaison with IBM's technical team, venue setup and equipment testing, participant registration system, and live session moderation. The workshop attracted 60–80 attendees from across the university's engineering departments. Facilitated hands-on coding segments where participants deployed their first cloud applications under guided supervision.
- ▶ **Edge-Core Hackathon:** Led a team of four to build HandSlash — an AI-powered touchless hand gesture recognition system using MediaPipe Tasks HandLandmarker with TFLite float16 model — in under 36 hours. The system enables camera-based gesture control for grabbing objects, clicking, and typing without physical contact. Won first place among 25 competing teams, recognized for technical innovation and clean implementation.

Academic Contributions

- ▶ **DZone Publication:** Authored a technical article on ORCA's MOGA approach for urban parking optimization, published on DZone — one of the largest developer communities with 5M+ monthly readers. The article bridges academic research and industry practice, explaining multi-objective optimization concepts in an accessible format for software engineers and smart city developers.
- ▶ **Research Collaboration:** Contributed to collaborative research with Aditi Singhal and Nithilan Rameshkumar (Amrita Vishwa Vidyapeetham) on the ORCA project. Co-developed the multi-objective genetic algorithm implementation and conducted statistical analysis using Kruskal-Wallis and Mann-Whitney U-tests to validate performance improvements over baseline parking allocation methods.

Publications Under Preparation

1. **ORCA: Optimized Routing and Car-park Assignment via Multi-Objective Genetic Algorithms**
Target Venue: IEEE Transactions on Intelligent Transportation Systems / ACM Computing Surveys

Status: Revised and resubmitted after first round of review

Co-authors: Aditi Singhal, Nithilan Rameshkumar, Murali K

Description: Extended version of the DZone paper with additional experiments on scalability analysis (up to 100,000 nodes), distributed GA implementations, and a comprehensive comparison with NSGA-III and Particle Swarm Optimization baselines. Includes a new section on real-world deployment challenges and IoT integration architecture.

2. **PRISM: A Probabilistic Real-time Intelligent Scheduling Mechanism for Minimizing Context Switch Overhead in Operating Systems**

Target Venue: IEEE ICCA / IEEE ICCCS

Status: Manuscript in preparation; simulation complete

Description: Extends traditional OS scheduling research by integrating Markov Chain probabilistic analysis with machine learning classification for process burst prediction. Includes novel Weighted Moving Average sequence prediction layer and online learning feedback loop for adaptive scheduling.

References

Academic Reference

Prof. Murali K
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Amrita Vishwa Vidyapeetham
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Industry Reference

Cellium Networks
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